SC2104/CE3002 Sensors & Interfacing
Tutorial 4 – DC motor and Encoder

1. A DC motor is fitted with two hall effect sensors based rotary encoder that utilizes a 24 poles magnetic disc (i.e., 12 North poles and 12 South poles) with quadrature output.

   a) Determine the resolution (in terms of angles) if we use
      i. X1 encoding
      ii. X2 encoding
      iii. X4 encoding

   if there are 24 poles, means 12 rising edge, 12 falling edge therefore 12 pulses per revolution. X1 encoding means every rising edge thus 360/12. X2 encoding means rising+falling edge, thus 360/24. X4 means B channel is 90 degree away thus is 360/48 resolution.

   b) Two such DC gear motors with 30:1 gear ratio are used to drive a two-wheel robotic vehicle.
      i. If the diameter of each wheel is 6cm, determine the distance travelled by the robot when the count value is equal to 1800 with an X2 encoding.
      ii. What is the number of count value needed to make the robot travel a 1m distance?
      iii. Comment on the resolution achieved in this system.

   no. of motor revolution = 1800/(360/24) = 75No. of wheel rev = (75/30)*1 due to gear ratio, then u multiply by circumference of wheel pi6.

   no. of wheel rev = 100/6pi. no. of gear rev = ans *30. count value = ans * 360/24

   smallest angle to be accurate with X2 encoding, 360/24. slowly calculate to the distance == 0.26cm